ECON 2050 – Mathematical Economics

Optimization I: Basics

Optimization problems

We are now in a position to deal with optimization problems!

In this course, we focus on optimization problems that the values of a given function $f : S \to \mathbb{R}$ are to be maximized or minimized over a set D, where $D \subseteq S \subseteq \mathbb{R}^n$.

- Call the function f the **objective function**, and the set D is called the **constraint set**.

We can describe a maximization problem as follows:

$$\text{maximize } f(\boldsymbol{x}) \quad \text{subject to } \boldsymbol{x} \in D;$$

or more compactly, $\max_{\boldsymbol{x} \in D} f(\boldsymbol{x})$. A minimization problem can be described analogously.

A **solution** to a maximization [minimization] problem is a point $\boldsymbol{x} \in D$ such that $f(\boldsymbol{x}) \geq f(\boldsymbol{y})$ $[f(\boldsymbol{x}) \leq f(\boldsymbol{y})]$ for all $\boldsymbol{y} \in D$.

As the following examples point out, in a given maximization problem, a solution may fail to exist; even if a solution exists, it need not be unique.

Example: Let $D = \mathbb{R}_+$ and $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x$. Consider the maximization problem $\max_{x \in D} f(x)$: we have $f(D) \equiv \{y \in \mathbb{R} : y = f(x) \text{ for some } x \in D\} = \mathbb{R}_+$, meaning that $f(x) \to \infty$ as $x \to \infty$, so this maximization problem has no solution.

Example: Let $E = [0, 1)$. Consider the maximization problem $\max_{x \in E} f(x)$, we have $f(E) = [0, 1)$. This means that the as x approaches 1, $f(x)$ get closer and closer to 1 but never reaches 1 since $1 \notin E$. Thus, this maximization problem has no solution.

Example: Let $C = [-1, 1]$ and $g : \mathbb{R} \to \mathbb{R}$ be defined by $g(x) = x^2$. By drawing a picture, we see that the maximization problem $\max_{x \in C} g(x)$ has two solutions: $x = -1$ and $x = 1$.

Local and global extrema

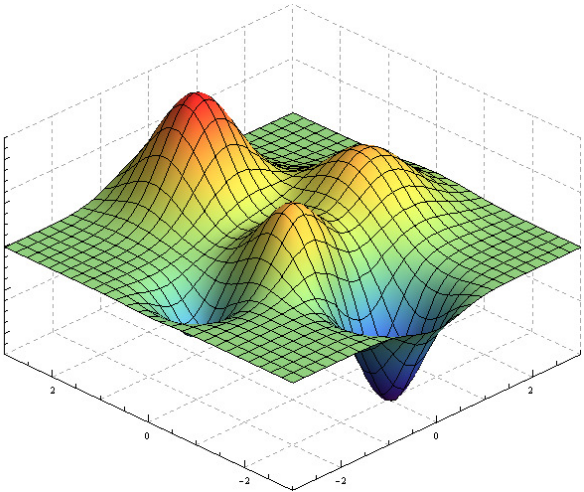
Definition

Let $D \subseteq \mathbb{R}^n$ and $f : D \to \mathbb{R}$. We say that $\boldsymbol{x}^* \in D$ is a

- **(global) maximum [minimum]** of f if $f(\boldsymbol{x}^*) \geq [\leq] f(\boldsymbol{x})$, $\forall \boldsymbol{x} \in D$.
- **(global) extremum** of f if $\boldsymbol{x}^*$ is a (global) minimum or a (global) maximum of f.
- **local maximum [minimum]** of f if there exists an open ball $B_r(\boldsymbol{x}^*)$ such that $f(\boldsymbol{x}^*) \geq [\leq] f(\boldsymbol{x})$, $\forall \boldsymbol{x} \in B_r(\boldsymbol{x}^*) \cap D$.
- **local extremum** of f if $\boldsymbol{x}^*$ is a local minimum or a local maximum of f.

Every (global) maximum [minimum] is a local maximum [minimum], but the converse is *false*.

Example: The function $f : [-3,3] \times [-3,3] \to \mathbb{R}$ with the graph depicted below has 1 global maximum, (at least) 3 local maxima, 1 global minimum and (at least) 2 local minima:



First-order necessary conditions for *interior* local extrema

First, we generalize the one-variable calculus necessary condition "$f'(x) = 0$:"

Definition

Let $D \subseteq \mathbb{R}^n$, and $f : D \to \mathbb{R}$ is a differentiable function. Then $\boldsymbol{x} \in \text{int}D$ is a **critical point** of f if

$$\nabla f(\boldsymbol{x}) = 0,$$

that is, if

$$\frac{\partial f(\boldsymbol{x})}{\partial x_1} = 0, \frac{\partial f(\boldsymbol{x})}{\partial x_2} = 0, \ldots, \frac{\partial f(\boldsymbol{x})}{\partial x_n} = 0.$$

Theorem (First-order necessary condition)

Let $D \subseteq \mathbb{R}^n$, and $f : D \to \mathbb{R}$ is a differentiable function. If $\boldsymbol{x} \in \text{int}\, D$ is a local extremum of f, then $\boldsymbol{x}$ is a critical point of f.

Proof sketch (two variable case):

Suppose $f : \mathbb{R}^2 \to \mathbb{R}$ achieves a local maximum at point (x_1^*, x_2^*). Then

$$\exists r > 0, \forall (x_1, x_2) \in B_r(x_1^*, x_2^*) : \quad f(x_1, x_2) \leq f(x_1^*, x_2^*)$$

Thus, if we keep $x_2 = x_2^*$ fixed and define $g : \mathbb{R} \to \mathbb{R}$ given by $g(x_1) \equiv f(x_1, x_2^*)$,

$$\forall x_1 \in pr_1(B_r(x_1^*, x_2^*)) : \quad g(x_1) = f(x_1, x_2^*) \leq f(x_1^*, x_2^*) = g(x_1^*)$$

This implies that the one variable function g has a local maximum at $x_1 = x_1^*$, which implies that $g'(x_1^*) = 0$ and therefore

$$\frac{\partial f(x_1^*, x_2^*)}{\partial x_1} = 0$$

Analogously, keeping $x_1 = x_1^*$ fixed, we obtain $\frac{\partial f(x_1^*, x_2^*)}{\partial x_2} = 0$.

Changing $\leq$ for $\geq$ demonstrates the same result for a local minimum.

Example: Consider $f : \mathbb{R}^2 \to \mathbb{R}, (x_1, x_2) \mapsto x_1^2 + x_2^2$.

To determine f's critical points we calculate the partial derivatives and equate them to zero:

$$\frac{\partial f(x_1, x_2)}{\partial x_1} = 2x_1 = 0$$
$$\frac{\partial f(x_1, x_2)}{\partial x_2} = 2x_2 = 0$$

Solving the system, we obtain the only critical point $(0, 0)$.

Because $\mathbb{R}^2$ is open (as any $\boldsymbol{x} \in \mathbb{R}^2$ is an interior point), we know from the previous theorem that every $(x_1, x_2) \neq (0, 0)$ is neither a local maximum nor a local minimum.

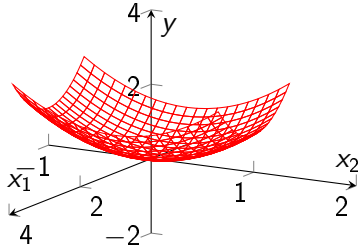
However, we do not know if $(0, 0)$ is a local maximum or a local minimum or neither.

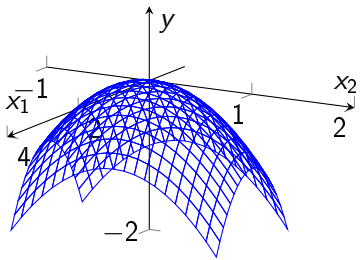
To sum up: If $D \subseteq \mathbb{R}^n$ is open and $f : D \to \mathbb{R}$ is differentiable,

non-critical points are not local extrema (and therefore not global extrema).

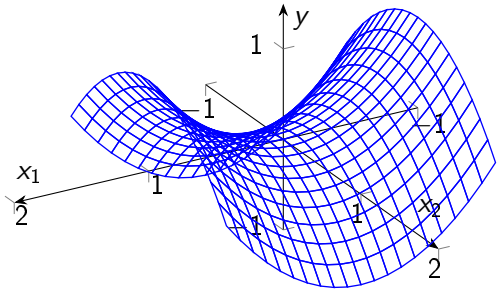
But: Not every critical point is say a local maximum, or even a local extremum. E.g., $(0, 0)$ is a critical point of all of the following functions (verify!), and

- a local minimum for $f : \mathbb{R}^2 \to \mathbb{R}, (x_1, x_2) \mapsto x_1^2 + x_2^2$
- a local maximum of $f : \mathbb{R}^2 \to \mathbb{R}, (x_1, x_2) \mapsto -(x_1^2 + x_2^2)$





- a **saddle point** (= a critical point that is not a local extremum) of $f : \mathbb{R}^2 \to \mathbb{R}, (x_1, x_2) \mapsto x_1^2 - x_2^2$



Sufficient conditions

Above, we discussed that a *necessary condition* for a point to be a *local* extremum is that it is a critical point.

- In general we don't even know whether a critical point is a local extremum, let alone global

The following theorem shows that being a critical point is *sufficient* for a *global* maximum [minimum] of *concave [convex]* functions.

Theorem (Sufficient conditions for global extrema)

Let $D \subseteq \mathbb{R}^n$ be convex, $f : D \to \mathbb{R}$ be a differentiable function. Then:

1. *Suppose f is concave, then $\boldsymbol{x}^*$ is a (global) maximum of f if it is a critical point of f.*
2. *Suppose f is convex, then $\boldsymbol{x}^*$ is a (global) minimum of f if it is a critical point of f.*

Theorem (Sufficient conditions for global extrema)

Let $D \subseteq \mathbb{R}^n$ be convex, $f : D \to \mathbb{R}$ be a differentiable function. Then:

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2. *Suppose f is convex, then $\boldsymbol{x}^*$ is a (global) minimum of f if it is a critical point of f.*

Proof: We only prove part 1; part 2 can be proved analogously. Since f is concave,

$$f(\boldsymbol{y}) - f(\boldsymbol{x}) \leq \nabla f(\boldsymbol{x}) \cdot (\boldsymbol{y} - \boldsymbol{x}), \quad \forall \boldsymbol{x}, \boldsymbol{y} \in D$$

(by the theorem on lecture 7 slide 26). If $\boldsymbol{x}^*$ is a critical point of f then $\nabla f(\boldsymbol{x}^*) = 0$, meaning that for all $\boldsymbol{x} \in D$,

$$f(\boldsymbol{x}) - f(\boldsymbol{x}^*) \leq 0 \cdot (\boldsymbol{x} - \boldsymbol{x}^*) = 0,$$

which is equivalent to $f(\boldsymbol{x}) \leq f(\boldsymbol{x}^*)$ for all $\boldsymbol{x} \in D$. Hence $\boldsymbol{x}^*$ is a global maximum of f.

Example: The partial derivatives of $f : \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto f(x, y) = x^2 + y^2$ are

$$\frac{\partial f(x, y)}{\partial x} = 2x,$$
$$\frac{\partial f(x, y)}{\partial y} = 2y$$

The unique critical point of f is $(0, 0)$. Since we know that f is convex (see, for example, Lecture 7 slide 30), $(0, 0)$ must be a global minimum of f.

Exercise: Determine the extrema of $f : \mathbb{R}^2 \to \mathbb{R}$ given by $f(x, y) = -x^2 + xy - y^2$. Classify them into local or global (you can use the fact that f is concave).

We calculate the critical points:

$$\frac{\partial f(x, y)}{\partial x} = -2x + y = 0$$
$$\frac{\partial f(x, y)}{\partial y} = -2y + x = 0$$

The only critical point is $(0, 0)$. Because $\mathbb{R}^2$ is open, this is the only candidate for local extrema. Since f is a concave function, $(0, 0)$ is a global maximum and therefore also a local maximum.

Example: Consider $f : \mathbb{R}^2_{++} \to \mathbb{R}$ given by $f(x_1, x_2) = \ln\left(x_1^2 + x_2^2\right)$.

To determine f's critical points we calculate the partial derivatives and equate them to zero:

$$\frac{\partial f(x_1, x_2)}{\partial x_1} = \frac{2x_1}{x_1^2 + x_2^2} = 0$$
$$\frac{\partial f(x_1, x_2)}{\partial x_2} = \frac{2x_2}{x_1^2 + x_2^2} = 0$$

We see that f has no critical points in $\mathbb{R}^2_{++}$ (note that $(0, 0) \notin \mathbb{R}^2_{++}$).

Then because $\mathbb{R}^2_{++}$ is open, f has neither local maxima nor local minima.

- Since every global maximum [minimum] is a local maximum [minimum], f has neither global maximum nor global minimum.

Existence of global extrema

We mentioned the following important result in Lecture 2:

Theorem (Extreme Value Theorem)

Let $D \subseteq \mathbb{R}^n$ be compact and $f : D \to \mathbb{R}$ be continuous. Then f has at least one global maximum and at least one global minimum. That is, there exist $\boldsymbol{x}^{\max}, \boldsymbol{x}^{\min} \in D$ such that

$$f(\boldsymbol{x}^{\max}) \geq f(\boldsymbol{x}) \geq f(\boldsymbol{x}^{\min}) \quad \forall \boldsymbol{x} \in D.$$

Proof idea:

- The key step is that the range of f, $f(D)$, is a compact subset of $\mathbb{R}$ provided that f is continuous and D is compact; thus, $f(D)$ is closed and bounded.
- Boundedness means that $\max f(D)$ and $\min f(D)$ are bounded away from infinity, and closeness implies that $\max f(D)$ and $\min f(D)$ can be attained.

Why do the conditions of the theorem matter?

Example: Let $C = \mathbb{R}$ and $f : C \to \mathbb{R}$ be given by $f(x) = x$.

Then f is continuous, but C is not compact (closed but not bounded). Since $f(C) = \mathbb{R}$, f has neither a (global) maximum nor a minimum.

Example: Let $D = (0, 1)$ and $f : D \to \mathbb{R}$ be given by $f(x) = x$.

Then f is continuous, but D is again non-compact (bounded but not closed). Since $f(D) = (0, 1)$, f has neither a (global) maximum nor a minimum. (Note that $x = 0$ and $x = 1$ are not in D.)

Properties of maximization/minimization problems

Proposition

Let $f : D \to \mathbb{R}$, where $D \subseteq \mathbb{R}^n$, and let $-f$ denote the function whose value at $\boldsymbol{x}$ is $-f(\boldsymbol{x})$. Then

- *$\boldsymbol{x}$ is a (global) maximum [minimum] of f if and only if $\boldsymbol{x}$ is a (global) minimum [maximum] of $-f$;*
- *$\boldsymbol{y}$ is a local maximum [minimum] of f if and only of $\boldsymbol{y}$ is a local minimum [maximum] of $-f$.*

The above result tells us that a generic minimization problem

$$\min_{x \in D} g(x)$$

is equivalent to

$$\max_{x \in D} -g(x);$$

in other words, every minimization problem may be represented as a maximization problem.

The following result is useful for calculating (global or local) extrema since it allows us to *simplify* the optimization problem.

Proposition

If $h : \mathbb{R} \to \mathbb{R}$ is an increasing function, $D \subseteq \mathbb{R}^n$, $g : D \to \mathbb{R}$ and $\mathbf{x}$ is a maximum [minimum] of g, then $\mathbf{x}$ is also a maximum [minimum] of $h \circ g$. If $h : \mathbb{R} \to \mathbb{R}$ is strictly increasing, the maxima [minima] of $h \circ g$ and g coincide.

Proof idea: compositing with an increasing function preserves maximality [minimality].

Example: Instead of minimizing

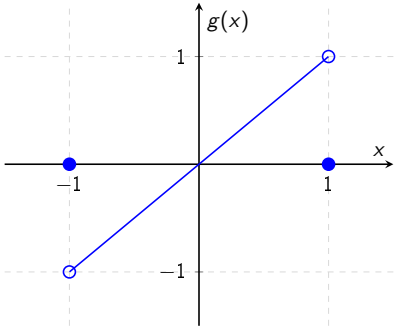
$$f(x, y) = e^{x^2+y^2},$$

we can minimize $g(x, y) = x^2 + y^2$ — since $h(z) = e^z$ is strictly increasing and $f = h \circ g$, the minima of f and g coincide.

Example: Let $E = [-1, 1]$ and $g : E \to \mathbb{R}$ be given by

$$g(x) = \begin{cases} 0, & \text{if } x = -1 \text{ or } x = 1; \\ x, & \text{if } -1 < x < 1. \end{cases}$$

Note that E is compact, but f is not continuous at -1 and 1. In this case, $g(E) = (-1, 1)$, and hence f has neither a (global) maximum nor a minimum.



Exercise: Check whether the conditions of the extreme value theorem hold, and find the (global) extrema if they exist.

1. $C = [-1, 1]$ and $f : C \to \mathbb{R}$ be given by $f(x) = x^2$.
   - Both conditions hold because C is closed and bounded, and hence compact, and f is continuou
   - By the extreme value theorem, a maximum and a minimum exist.
   - $x = -1$ and $x = 1$ are global maxima and $x = 0$ is a global minimum.

2. $C = [-1, 1]$ and $g : C \to \mathbb{R}$ be given by

$$g(x) = \begin{cases} x & \text{if } -1 \leq x \leq 0, \\ 1 - x^2 & \text{if } 0 < x \leq 1. \end{cases}$$

   - C is compact but g is discontinuous at $x = 0$; the extreme value theorem doesn't apply.
   - There is no global maximum, but $x = -1$ is a global minimum.

Caveats:

- The theorem does not say where these extrema are or how to find them, it only says that these points exist.
- Importantly, the conditions of the theorem are not necessary: if one or more of the conditions fails, maxima and minima may still exist.

Example: Let $D = [-1, \infty)$, and let $f : D \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x & \text{if } -1 \leq x < 0, \\ 1 & \text{if } x \geq 0. \end{cases}$$

We see that f is not continuous at $x = 0 \in D$, and D is not compact (closed but not bounded); but any $x \geq 0$ is a global maximum of f, and $x = -1$ is a global minimum.

A (not-always-useful) cookbook procedure for finding (global) extrema

Let S be a compact subset of $\mathbb{R}^n$, and let $f : S \to \mathbb{R}$ be continuous. Then the extrema of f can be found as follows:

1. Find all critical points in int$S$.
2. Among the points in $S \backslash \text{int}S$ (i.e., the boundary points), find those at which the value of f is largest.
3. Among all the points you have found in the previous two steps, the ones at which the value of f are the largest [smallest] are the (global) maxima [minima].

This approach is not always useful, because evaluating all boundary points can be difficult.

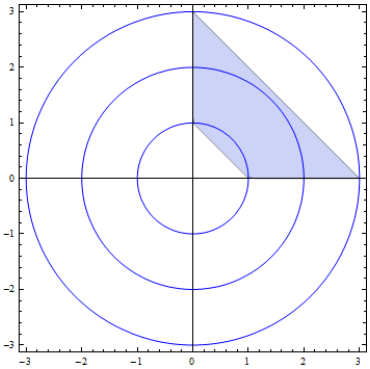
- In the next three lectures, we will discuss a simpler and more systematic approach.
- However, in certain cases, this approach can be handy.

Example: Consider the minimization problem $\min_{(x,y)\in S} f(x,y)$, where $f : \mathbb{R}^2 \to \mathbb{R}$ is given by $f(x,y) = x^2 + y^2$, and

$$S = \left\{(x,y) \in \mathbb{R}^2 : 1 \leq x + y \leq 3, x \geq 0, y \geq 0\right\}.$$

- We know from slide 13 that no interior point of S is a solution, because the unique critical point $(0,0)$ is not in S.
- However, by the extreme value theorem, the minimization problem has a solution.
- So: Which of the boundary points are solutions?

Below, we depict $S = \{(x, y) \in \mathbb{R}^2 : 1 \leq x + y \leq 3, x \geq 0, y \geq 0\}$ and some level curves of f.



The graph shows that the unique solution of the minimization problem is $(\frac{1}{2}, \frac{1}{2})$.

In fact, we can say more: $(3, 0)$ and $(0, 3)$ maximize f on S.